Machine Learning-Based Classification for Learning Outcome Evaluation in Online Learning

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***Abstract*— Learning outcome evaluation in online learning environments is frequently operationalized through final grades or aggregated assessment scores, resulting in a limited representation of student competency within Outcome-Based Education (OBE) frameworks. This study examines how learning outcomes are evaluated using assessment data in online learning and investigates the potential of machine learning-based classification to support learning outcome evaluation. An implementation-based methodology was applied to analyze student assessment records using several classification models, including Logistic Regression, Decision Tree, K-Nearest Neighbors, and Random Forest. Among the evaluated models, the Random Forest approach achieved the most consistent classification performance and F1-score, indicating that ensemble-based methods provide more stable and balanced classification results. The findings suggest that while machine learning techniques can effectively analyze assessment data, learning outcome evaluation remains constrained when outcomes are equated solely with grades. These results highlight the limitations of grade-oriented evaluation and demonstrate that machine learning-based analytical approaches can support more systematic interpretation of learning outcome attainment. This approach provides practical implications for educators and institutions in enhancing curriculum evaluation, learning analytics, and quality assurance within online learning environments.**

Keywords—Learning outcome evaluation, outcome-based education, online learning, machine learning, learning analytics.

I. INTRODUCTION

Online learning is already a crucial part of higher education, particularly following the accelerated digital transformation triggered by global disruptions. The widespread adoption of Learning Management Systems, virtual classrooms, and digital assessment platforms has enabled institutions to deliver instruction at scale while maintaining the continuity of teaching and learning processes. As online learning environments continue to expand, ensuring the quality and effectiveness of education has become a major issue, especially in relation to whether students genuinely achieve the intended learning outcomes [1], [2].

Within the Outcome-Based Education (OBE) paradigm, learning outcomes serve as explicit statements of the competencies that learners are expected to demonstrate upon completing a course or program. These outcomes are not limited to knowledge acquisition but encompass cognitive, affective, and psychomotor domains that collectively reflect student competence and readiness [3]. Consequently, evaluating learning outcomes is not merely a procedural requirement but a critical mechanism for ensuring educational accountability, curriculum relevance, and alignment with program educational objectives. Inaccurate or superficial evaluation of learning outcomes may lead to misleading conclusions regarding student competence, ultimately weakening curriculum improvement efforts and institutional quality assurance processes [4], [5].

Despite this importance, learning outcome evaluation in online learning environments is commonly operationalized through final grades or partial assessment scores such as quizzes, midterm examinations, and final examinations. Grades provide a convenient and standardized representation of student performance; however, evaluating learning outcomes solely through aggregated scores presents a narrow and potentially misleading view of student achievement [6]. In OBE-oriented education, learning outcomes are designed to capture multidimensional competencies, yet grade-centric evaluation practices tend to emphasize cognitive performance while underrepresenting affective factors such as motivation and engagement, as well as psychomotor skills that may be implicitly demonstrated through learning activities [2], [3]

This limitation is further reflected in prior studies within learning analytics and educational data mining, where quantitative approaches predominantly focus on predicting academic success using final grades as the primary indicator [7], [8] While such methods offer valuable insights into performance trends, they often fail to account for the complexity of learning outcomes in online learning contexts. Student interaction, participation patterns, and learning behaviors—elements that play a crucial role in competency development—are frequently overlooked, resulting in incomplete representations of learning achievement [9], [10]

The challenge of learning outcome evaluation becomes more pronounced in online learning environments, where learning processes are less directly observable than in traditional face-to-face circumstances. In the absence of comprehensive evaluation mechanisms, learning outcomes are frequently inferred from assessment results rather than from the learning processes that contribute to competency formation. As a result, students who obtain satisfactory grades may not fully demonstrate all intended learning outcomes, while learners who exhibit meaningful engagement and learning effort may not be adequately recognized [9], [11]

Recent research has highlighted the need for more holistic approaches to learning outcome evaluation and has emphasized the shortcomings of grade-based assessment

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models [12], [13]. However, due to institutional constraints and limitations in data utilization, many evaluation practices continue to rely on numerical grades as proxies for learning outcomes, despite growing awareness of their insufficiency [14] This gap indicates the necessity for alternative evaluation models that remain practical within existing online learning infrastructures while aligning more closely with the principles of outcome-based education.

Accordingly, this study addresses the following research questions:
RQ1: How can machine learning-based classification support learning outcome evaluation in online learning environments?
RQ2: How can assessment data be interpreted to represent learning outcome attainment within grade-based evaluation contexts?

In this context, machine learning offers promising opportunities to enhance learning outcome evaluation by enabling the analysis of assessment data beyond surface-level scores [15]. By identifying patterns within quiz and examination results, machine learning can provide a more nuanced interpretation of learning outcome attainment [16]. This study addresses the limitation of grade-centered evaluation by proposing a machine learning-based approach to support the interpretation of multidimensional learning outcomes within OBE-oriented online learning. Despite the extensive use of learning analytics, existing approaches predominantly focus on grade prediction rather than explicit learning outcome evaluation, highlighting a critical research gap.

II. Literature Review

Early warning systems that identify at-risk learners, engagement analysis, and student performance prediction are just a few of the many educational applications of machine learning. These methods usually rely on information from learning management systems, including interaction logs and test results. However, with little focus on explicit evaluation in line with outcome-based education frameworks, the majority of research operationalize learning outcomes primarily through grades.

Several studies have investigated learning outcome evaluation in higher education through the use of learning analytics systems. A systematic literature review reported in Learning Outcomes Evaluation Through Learning Analytics Systems in Higher Education indicates that most existing learning analytics frameworks predominantly rely on quantitative indicators such as grades, assessment scores, and course completion rates extracted from Learning Management Systems (LMS). Although these approaches support institutional decision-making and early detection of learners who are at risk, learning outcomes are often operationalized in a generalized manner, equating achievement with academic performance indicators. The review further highlights that many learning analytics implementations emphasize system usability and data visualization for instructors, while providing limited insight into whether students have achieved the intended learning outcomes across broader competency domains. This suggests that evaluation practices are frequently shaped by data availability and implementation convenience rather than by comprehensive representations of learning outcome attainment in online learning environments [6]

Another stream of studies applies learning analytics to examine student engagement as an indirect indicator of learning improvement. Prior studies analyze behavioral indicators derived from Learning Management System (LMS) interactions, including login frequency, content access, and participation in online learning activities, to identify engagement patterns. The results indicate a strong relationship between engagement indicators and academic performance, allowing early identification of potential learning difficulties. However, learning outcome evaluation remains a secondary focus, as engagement metrics are predominantly utilized to predict grades or course completion rather than to assess the attainment of specific learning outcomes. As a result, although engagement-based analytics enhance the monitoring of student behavior, their alignment with Outcome-Based Education (OBE) principles remains limited, and the conceptual linkage between engagement indicators and multidimensional learning outcomes is insufficiently addressed [17].

Research on online assessment practices in higher education further illustrates how learning outcome evaluation is often driven by considerations of efficiency and scalability. The study reported in *Online Assessment in Higher Education* reviews various assessment methods used in online learning, including quizzes, assignments, tests, and summative examinations. The findings indicate that these practices are primarily designed to support efficient administration and automated grading, particularly in large-scale courses. Consequently, learning outcomes are frequently inferred from aggregated assessment scores rather than evaluated in terms of competency representation, reinforcing the persistence of grade-oriented evaluation models and the challenge of aligning online assessment practices with the multidimensional learning outcomes emphasized within OBE frameworks [18]. Assessment scores are widely available in online learning environments; however, their use is typically limited to summative grading rather than being systematically analyzed to reflect specific learning outcome attainment [19].

Overall, existing research demonstrates that while learning analytics and online assessment systems enhance efficiency, monitoring, and predictive capabilities, learning outcome evaluation in online learning remains predominantly grade-centric and indirectly inferred. Engagement indicators and assessment scores are frequently treated as proxies for learning outcomes, resulting in limited alignment with the multidimensional competencies defined in Outcome-Based Education [20]. These limitations highlight the need for an evaluation approach that explicitly links assessment data to learning outcome attainment and leverages analytical techniques capable of supporting outcome-oriented interpretation within existing online learning infrastructures.

Despite the growing adoption of learning analytics and engagement-based indicators, existing studies predominantly employ analytical techniques for performance prediction rather than for explicit learning outcome evaluation. Machine learning methods are often utilized to forecast grades, retention, or course completion, while their potential to support multidimensional learning outcome assessment remains underexplored. This indicates a methodological gap between the availability of assessment data and its use for outcome-oriented evaluation aligned with OBE principles.

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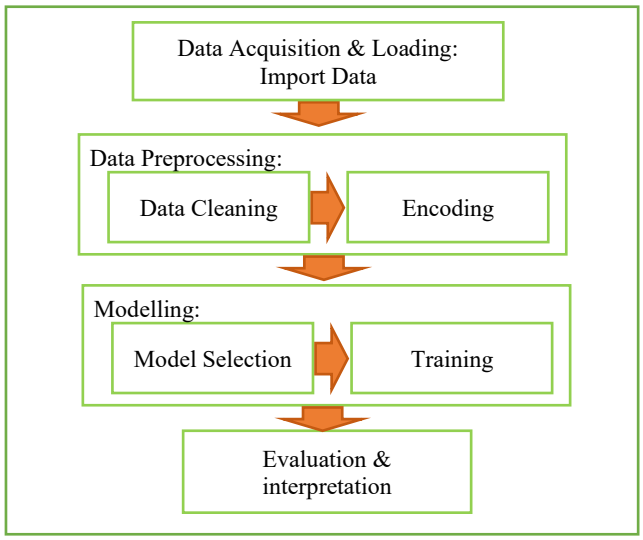


Fig. 1. Research Stages

III. Research Methodology

This study employed an implementation-based research methodology to examine learning outcome evaluation in an online learning environment using existing assessment data. Rather than treating grades as direct representations of learning outcomes, assessment results were utilized as observable indicators to support the interpretation of learning outcome attainment within an Outcome-Based Education (OBE) framework. The analytical process was implemented using Google Colab to ensure reproducibility and consistency across data processing and model execution stages.

The methodology began with data acquisition and loading, where academic performance records were imported from cloud-based spreadsheet storage into the analysis environment. The dataset contained student assessment results derived from quizzes and examinations commonly used in online learning contexts. Initial data inspection was conducted to verify completeness, structural consistency, and suitability for analytical processing.

513 student records from seven courses in an online learning environment make up the dataset. Individual student assessment results are represented by each record. Grading categories based on final scores, which were the goal variable for classification, were used to operationalize learning outcome attainment. The dataset is openly accessible on Zenodo with the link below https://doi.org/10.5281/zenodo.19871067

TABLE I. Dataset Variable And Type

Variable	Explanation	Type
IDS	Student ID number	Numeric
Course	Material studied by students in a class scheduled	Char
Learning Outcomes	Expected knowledge and skills students gain after completing the course	Char
Individual Assignment	Individual assessment to measure academic achievement	Numeric
Group Assignment	Evaluation aimed at measuring team achievement	Numeric
Quiz	Online short assessment conducted at a designated time	Numeric
Final Exam	Final examination to assess overall learning outcomes	Numeric
Final Score	The cumulative score derived from all components	Numeric
Grading	Student grading categories derived from the final score range	Char

Data preprocessing was subsequently performed to prepare the dataset for machine learning analysis. This phase involved encoding categorical variables, validating data types, and standardizing attribute names to improve clarity and consistency. Missing data, if present, were handled using removal, ensuring dataset consistency prior to model training. Categorical assessment outcomes were transformed into numerical representations to enable compatibility with classification models, ensuring that the dataset accurately reflected structured assessment indicators. To provide further clarity on these transformations, Table II presents a representative sample of the student assessment records before and after the preprocessing and feature mapping stages.

TABLE II. Example of Data Transformation during Preprocessing

Stage	Column Name	Column Value
Raw Data	Individual_Assignment1	86
	Individual_Assignment2	83
	Group_Assignment1	97
	Group_Assignment2	85
	Group_Assignment3	97
	Quiz1	100
	Quiz2	90
	Final_Exam	81
	Grading	A
Preprocessed Data	X2	86
	X3	83
	X4	97
	X5	85
	X6	97
	X7	100
	X8	90
	X9	81
	Y	4

Classification models were chosen and put into use after preprocessing in order to examine trends in learning outcome achievement based on assessment data. The models that were chosen represented both linear and non-linear methods of classification and included Random Forest, Decision Trees, K-Nearest Neighbors, and Logistic Regression. In order to ensure that the results could be effectively interpreted within an Outcome-Based Education framework, interpretability, computational efficiency, and suitability for organized

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educational data were given top priority during the model selection process. Instead of concentrating only on predicted accuracy or student performance estimation, this strategy supports the study's goal of assisting learning outcome evaluation by discovering classifying patterns.

TABLE III. MACHINE LEARNING MODEL PARAMETERS

Model Type	Algorithm	Parameter	Configuration
Baseline Models	Logistic Regression	Solver	liblinear
		Random State	42
	Decision Tree	Criterion	Gini Impurity
		Splitter	Best
	K-Nearest Neighbors	N-Neighbors	5 (k=5)
		Metric	Minkowski
Optimized Model	Random Forest Classifier	N-Estimators	200 Trees
		Max Depth	10
		Max Features	log2
		Random State	42

Model training involved fitting the selected classifiers to labeled outcome indicators, followed by parameter tuning to enhance classification stability and consistency. While Logistic Regression, Decision Tree, and K-Nearest Neighbors were implemented using their respective default parameters to establish a performance baseline, systematic hyperparameter tuning was specifically applied to the Random Forest model due to its superior potential in capturing complex assessment patterns. The tuning process for the Random Forest classifier focused on optimizing the ensemble structure, specifically increasing the number of estimators to 200 and limiting the maximum depth to 10 to balance predictive sensitivity and model stability. Table III details the specific parameters utilized for the baseline models and the optimized configuration for the Random Forest model. The resulting classifications were then analyzed to examine learning outcome patterns reflected through assessment data. Descriptive evaluation techniques were applied to observe classification distributions and to interpret how learning outcomes were operationalized within the proposed analytical framework. The findings derived from this process are presented and discussed in the subsequent section.

IV. RESULTS AND DISCUSSION

The dataset was utilized to develop and assess the classification model by assigning a larger portion of the data for training and the remainder for validation. Specifically, 80% of the records were allocated for model development, while 20% were reserved for performance evaluation. The confusion matrix illustrated provides a detailed verification of the model's classification accuracy. The results demonstrate that the Tuned Random Forest Classifier achieved nearly perfect identification for students in 'A' and 'D' categories, with 20 out of 23 'A' students and the only 'D' student correctly classified. Consistent with the reported accuracy of 0.83, the majority of classification errors were localized between adjacent categories—specifically between 'A' and 'B', and 'B' and 'C'. This pattern indicates that while the model effectively distinguishes broad performance tiers, the inherent numerical overlap in assessment scores creates marginal challenges in differentiating between closely related competency levels. From an Outcome-Based Education (OBE) standpoint, the high precision in identifying extreme performance categories (A and D) ensures that institutions can reliably identify both high-achieving students and those requiring immediate intervention

TABLE IV. CONFUSION MATRIX FOR TUNED RANDOM FOREST CLASSIFIER

True Label	**D**	1	0	0	0
	C	0	1	1	0
	B	0	0	7	2
	A	0	0	3	20
		D	**C**	**B**	**A**
		Predicted Label			

The following indicators have been selected to assess the performance of classification models:

- Accuracy

Accuracy represents the ratio of correctly classified instances to the total number of observations, providing an overall measure of classification correctness. In the context of educational data analysis, this metric indicates how well the model captures general learning outcome categories based on assessment data.

$$Accuracy = \frac{(TP+TN)}{(TP+FP+FN+TN)} \quad (1)$$

- Recall

Recall indicates the extent to which instances belonging to a positive class are correctly recognized by the classification model. Within this study, recall is particularly important for ensuring that students assigned to a given learning outcome category are accurately captured and not misclassified as belonging to other categories.

$$Recall = \frac{TP}{(TP+FN)} \quad (2)$$

- Precision

Precision refers to the proportion of instances predicted as positive that are correctly classified as positive. This metric evaluates the model's ability to minimize false positive classifications by indicating how reliable positive predictions are.

$$Precision = \frac{TP}{(TN+FP)} \quad (3)$$

- F1 Score

The F1-score offers a balanced evaluation by integrating precision and recall and is well suited for analyzing learning outcome categories that exhibit uneven class distributions.

$$F1\ Score = \sqrt{Recall} * Precision \quad (4)$$

The misclassification rate is the proportion of samples that were misclassified in several cases. Its value can be determined by following these steps:

$$Error = \frac{FP+FN}{(TP+TN+FP+FN)} \quad (5)$$

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TABLE V. PERFORMANCE METRICS OF CLASSIFICATION MODELS

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.74	0.70	0.74	0.72
Decision Tree Classifier	0.71	0.74	0.71	0.72
K-Nearest Neighbors	0.8	0.76	0.8	0.78
Random Forest Classifier	0.83	0.85	0.83	0.83

In the table above, the Random Forest Classifier achieves the highest accuracy (0.83), indicating the best overall performance among the evaluated models, while the Decision Tree Classifier shows the lowest accuracy (0.71). The K-Nearest Neighbors model also demonstrates relatively high accuracy (0.80), followed by Logistic Regression with an accuracy of 0.74.

In evaluating the reliability of positive outcome classification, the Random Forest approach demonstrated superior precision (0.85), compared to K-Nearest Neighbors (0.76), Decision Tree (0.74), and Logistic Regression (0.70). Higher precision in this context suggests improved reliability in identifying students who genuinely meet particular learning outcome criteria, which is critical for outcome-oriented evaluation in online learning environments.

When considering recall, both the Random Forest Classifier and K-Nearest Neighbors achieve high recall scores of 0.83 and 0.80, indicating their stronger ability to correctly identify positive cases. In contrast, Logistic Regression and Decision Tree Classifier record lower recall values of 0.74 and 0.71, respectively, reflecting weaker sensitivity in capturing all relevant instances.

Finally, the Random Forest Classifier recorded the strongest F1-score (0.83), demonstrating a well-balanced trade-off between precision and recall. The K-Nearest Neighbors model follows with an F1-Score of 0.78, while Logistic Regression and Decision Tree Classifier share the lowest F1-Score value (0.72), indicating comparatively less balanced performance.

The findings show that ensemble-based classification models—in particular, the Random Forest Classifier—are better at identifying recurring patterns in assessment data related to the achievement of learning objectives. Rather than focusing solely on predictive accuracy, these findings suggest that such models provide a more stable representation of learning outcome categories derived from assessment indicators.

To illustrate the integration of feature processing and model performance, Table VI presents selected modeling data inputs (X_n) alongside their corresponding prediction results. These examples demonstrate how the Random Forest Classifier interprets multidimensional assessment indicators to assign a final competency grade. The alignment between 'Actual Grading' and 'Predicted Grading' in these samples confirms the model's ability to capture structured learning outcome patterns from assessment data.

TABLE VI. EXAMPLES OF RANDOM FOREST CLASSIFIER OUTPUTS

X2	X3	X4	X5	X6	X7	X8	X9	Actual Grading	Predicted Grading
80	0	93	80	71	50	0	50	D	D
96	99	98	91	83	100	100	63	B	B
80	98	100	93	92	90	100	76	A	A

From an Outcome-Based Education perspective, this classification-based analysis supports a shift away from direct grade interpretation toward structured learning outcome evaluation. While assessment scores remain quantitative by nature, the application of machine learning enables the identification of patterns that reflect outcome attainment levels more systematically than raw grades alone. This aligns with prior studies that highlight the limitations of grade-centric evaluation models in online learning environments.

Furthermore, the results reinforce observations from the literature that conventional online assessment practices prioritize efficiency and measurability over competency representation. By leveraging classification models, assessment data can be reinterpreted to support learning outcome evaluation without requiring changes to existing online learning infrastructures. This approach addresses the gap identified in previous research, where learning analytics tools were primarily used for performance prediction rather than for explicit learning outcome evaluation aligned with OBE principles.

Overall, the findings demonstrate that machine learning-based classification can serve as a complementary analytical approach for learning outcome evaluation in online learning. Instead of replacing existing assessment mechanisms, the proposed method enhances the interpretability of assessment data, offering institutions a practical means to evaluate learning outcomes more comprehensively within current digital learning environments.

V. CONCLUSION

This study confirms that learning outcome evaluation in online learning environments remains predominantly grade-centric, despite the emphasis of Outcome-Based Education (OBE) on multidimensional competency attainment. The literature review and experimental findings demonstrate that assessment scores are frequently operationalized as proxies for learning outcomes, thereby limiting their ability to represent holistic student competencies.

The results further indicate that machine learning-based classification models can effectively analyze assessment data and reveal structured patterns related to learning outcome attainment. Rather than replacing existing assessment mechanisms, the proposed approach enhances the interpretability of grade-based data by supporting learning outcome evaluation within current online learning infrastructures.

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By highlighting the conceptual limitations of grade-oriented evaluation and demonstrating the potential of machine learning to support outcome-oriented interpretation, this study contributes to the development of more aligned learning outcome evaluation practices in online learning. Future research may extend this approach by incorporating additional learning indicators beyond assessment scores to further strengthen the evaluation of multidimensional learning outcomes in OBE-oriented educational settings.

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All authors contributed to this study. Avero Vincent Sukhitatha was responsible for data acquisition, preprocessing, and implementation of classification models including Logistic Regression, Decision Tree, K-Nearest Neighbors, and Random Forest, conducted the experimental analysis and drafted the initial manuscript. Meta Amalya Dewi contributed to the research conceptual framework, methodological design aligned with Outcome-Based Education (OBE), and supervised the overall study, performed critical review, interpretation of results, and manuscript refinement. All authors approved the final manuscript.

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